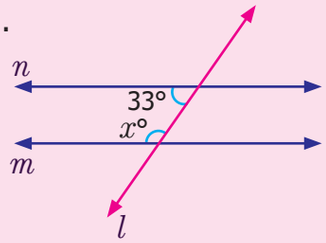
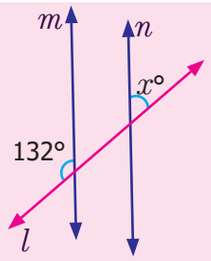
Try these

Find the value of x .





Example 5.8

(i) Name the angle that corresponds to $\angle 1$.
(ii) Name the angle that is alternate interior to $\angle 3$.
(iii) Name the angle that is alternate exterior to $\angle 8$.
(iv) Name the angle that corresponds to $\angle 8$.
(v) Name the angle that is alternate exterior to $\angle 7$.
(vi) Name the angle that is alternate interior to $\angle 6$.

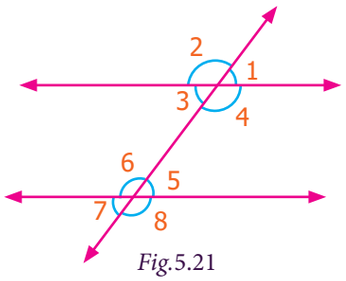


Fig.5.21

Solution

(i) The angle that corresponds to $\angle 1$ is $\angle 5$
(ii) The angle that is alternate interior to $\angle 3$ is $\angle 5$
(iii) The angle that is alternate exterior to $\angle 8$ is $\angle 2$
(iv) The angle that corresponds to $\angle 8$ is $\angle 4$
(v) The angle that is alternate exterior to $\angle 7$ is $\angle 1$
(vi) The angle that is alternate interior to $\angle 6$ is $\angle 4$

Example 5.9

(i) Which angles are corresponding angles to $b°$?
(ii) What is the measure of $b°$?
(iii) Which angles have the measure 68°?
(iv) Which angles have the measure 112°?

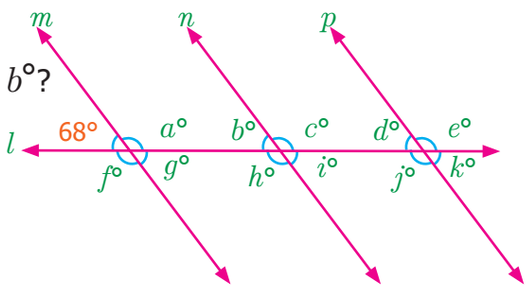


Fig.5.22

Solution

(i) The corresponding angles to $b°$ are $d°$ and 68°.
(ii) The measure of $b°$ is 68° (Because $b°$ corresponds to 68°)
(iii) The angles having measure 68° are $b°$, $d°$, $g°$, $i°$ and $k°$.
(iv) The angles having measure 112° are $a°$, $c°$, $e°$, $f°$, $h°$ and $j°$.

Example 5.10 If l is parallel to m, find the measure of x and y in the figure.

Solution Given l is parallel to m and n is transversal to l and m.

We get, $y = 2x$ [Vertically opposite angles are equal]

$y + 4x = 180°$ [sum of interior angles that lie on the same side of the transversal]

$2x + 4x = 180°$ [since $y = 2x$]

$6x = 180°$

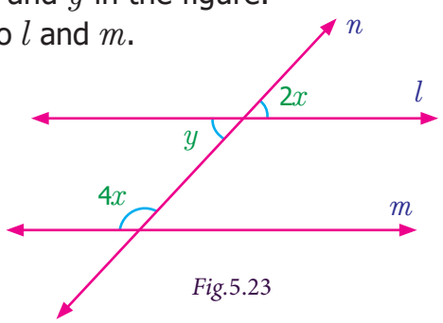


Fig.5.23

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